Probability Theory

Foundations for Financial Derivatives Pricing

Course Notes — Financial Derivatives Pricing

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- Probability Spaces
- σ-Algebras & Borel Sets
- Expectation & Conditioning
- Brownian Motion
- Martingales
- Axioms of Probability
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References: Shreve (2005); Morters & Peres (2010); Papanicolaou (2017)

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1 Why Probability?

Probability theory is ubiquitous in modern life. We encounter probabilistic reasoning whenever we hear statements such as:

- "There is a high chance it will rain tomorrow."
- "Party X is more likely to form the government."
- A doctor assesses the probability of success of various treatments.
- A bank computes the probability that a loan will be repaid.
- A pharmaceutical company calculates the probability of harmful drug side effects.

From time immemorial, human beings have been fascinated by the notion of looking into the future — probably because such a capacity would bring them certain advantages. Aristotle noted *"The probable is what usually happens."* Descartes argued that *"when it is not in our power to determine the most true opinions, we ought to follow the most probable."*

1.1 Brief History of Probability

1.1.1 Early Pioneers (16th–17th Century)

The formal study of probability began with problems in gambling and games of chance:

- **Cardan (1501–1576):** Wrote the first systematic treatment of probability in *Liber de Ludo Aleae*.
- **Kepler (1571–1630):** Applied probabilistic ideas in astronomy.
- **Galileo (1564–1642) & Merton:** Analysed dice problems systematically.
- **Pascal (1623–1662) & Fermat (1601–1665):** Solved *de Méré's Problem* by correspondence, founding the modern theory of probability.

The Chevalier de Méré observed from experience that:

$P(\text{at least one ace in 4 rolls of a single die}) > P(\text{at least one double-ace in 24 rolls of two dice})$.

He asked Pascal to verify this. The exact calculations are:

$$P_1 = 1 - \left(\frac{5}{6}\right)^4 \approx 0.5177, \qquad P_2 = 1 - \left(\frac{35}{36}\right)^{24} \approx 0.4914.$$

Indeed $P_1 > P_2$, confirming de Méré's intuition.

1.1.2 Modern Probability (20th Century)

- **Kolmogorov (1903–1987):** Connected the ideas of Borel and Lebesgue measure theory to probability, establishing its rigorous axiomatic foundation (1933).
- **Paul Lévy (1886–1971):** Set the tone for modern probability with seminal work on Stochastic Processes, Characteristic Functions, and Limit Theorems.

2 Probability Space

2.1 Sample Space and Events

In a random experiment, the **sample space** Ω is the collection of all possible outcomes. The outcomes must satisfy:

1. **Mutually exclusive:** No two outcomes can occur simultaneously.
2. **Collectively exhaustive:** Every possible outcome is included in Ω.

Example 2.1. Some common sample spaces:

- Coin toss: $\Omega = \{H, T\}$.
- Die roll: $\Omega = \{1, 2, 3, 4, 5, 6\}$.
- Angle of rotation of a wheel: $\Omega = [0, 2\pi]$.

An **event** is a property that can be observed to either hold or not hold after the experiment. Formally, an event is a subset $A \subseteq \Omega$. Key set-theoretic operations on events:

- **Union** ($A \cup B$): Event "A or B" occurs.
- **Intersection** ($A \cap B$): Event "A and B" occur.
- **Complement** (A^c): The contrary event "A does not occur".
- **Impossible event**: The empty set $\emptyset$.
- **Elementary event**: A singleton $\{\omega\}$ for $\omega \in \Omega$.

3 Axioms of Probability

3.1 Algebra and σ-Algebra

Before defining probability, we need a suitable collection of events on which probability is to be defined.

Let Ω be a non-empty set. A collection $\mathcal{F}$ of subsets of Ω is called an **algebra** if:

(i) $\Omega \in \mathcal{F}$.

(ii) If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$ (closed under complements).

(iii) If $A_1, A_2, \ldots, A_n \in \mathcal{F}$, then $\bigcup_{i=1}^n A_i \in \mathcal{F}$ (closed under finite unions).

Example 3.1. Two simple algebras on any Ω:

- $\mathcal{F} = \{\emptyset, \Omega\}$ (the trivial algebra).
- $\mathcal{F} = \{\emptyset, \Omega, A, A^c\}$ for any $A \subseteq \Omega$.

A collection $\mathcal{F}$ of subsets of Ω is called a **σ-algebra** (or σ-field) if:

(i) $\Omega \in \mathcal{F}$.

(ii) If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$.

(iii) If $A_1, A_2, A_3, \ldots \in \mathcal{F}$, then $\bigcup_{n=1}^\infty A_n \in \mathcal{F}$ (closed under *countable* unions).

The key distinction is that an algebra requires closure under *finite* unions, while a σ-algebra requires closure under *countably infinite* unions. Every σ-algebra is an algebra (take $A_{n+1} = A_{n+2} = \cdots = \emptyset$), but the converse is false.

Example 3.2 (An algebra that is not a σ-algebra)**.** Let $\Omega = \mathbb{N}$ (the natural numbers). Let $\mathcal{F}$ be the family of all subsets of $\mathbb{N}$ that are either *finite* or have a *finite complement*.

$\mathcal{F}$ is an algebra: If $A, B \in \mathcal{F}$, then all four cases of finiteness of A, A^c, B, B^c lead to $A \cup B$ being either finite or having a finite complement.

$\mathcal{F}$ is not a σ-algebra: Let $A_n = \{2, 4, 6, \ldots, 2n\} \in \mathcal{F}$. Then

$$B := \bigcup_{n=1}^\infty A_n = \{2, 4, 6, 8, \ldots\}$$

is infinite and $B^c = \{1, 3, 5, \ldots\}$ is also infinite, so $B \notin \mathcal{F}$.

Example 3.3 (A σ-algebra generated by intervals)**.** Let $\Omega = [0, 1]$ and suppose $\left[\frac{1}{n+1}, \frac{1}{n}\right] \in \mathcal{F}$ for all $n = 1, 2, \ldots$. Then:

(a) $\{0\} = [0, 1] \setminus \bigcup_{n=1}^\infty \left[\frac{1}{n+1}, \frac{1}{n}\right] \in \mathcal{F}$.

(b) $\left\{\frac{1}{n} : n \geq 2\right\} = \bigcup_{n=2}^\infty \left(\left[\frac{1}{n+1}, \frac{1}{n}\right] \cap \left[\frac{1}{n}, \frac{1}{n-1}\right]\right) \in \mathcal{F}$.

(c) $\left(\frac{1}{n}, 1\right] = \bigcup_{i=1}^{n-1} \left[\frac{1}{i+1}, \frac{1}{i}\right] \setminus \left[\frac{1}{n+1}, \frac{1}{n}\right] \in \mathcal{F}$.

(d) $\left(0, \frac{1}{n}\right] = \bigcup_{i=n}^\infty \left[\frac{1}{i+1}, \frac{1}{i}\right] \in \mathcal{F}$.

3.2 Generated σ-Algebra

Let Ω be a set and $\mathcal{B}$ a non-empty collection of subsets of Ω. The **σ-algebra generated by $\mathcal{B}$**, denoted $\sigma(\mathcal{B})$, is the *smallest* σ-algebra containing all sets of $\mathcal{B}$.

Remark 3.4. The generated σ-algebra always exists because 2^Ω (the power set) is a σ-algebra containing $\mathcal{B}$, and the intersection of any family of σ-algebras is again a σ-algebra.

Example 3.5.
- $\sigma(A) = \{\emptyset, \Omega, A, A^c\}$.
- $\sigma(A, B) = \{\emptyset, \Omega, A, A^c, B, B^c, A \cap B, A \cup B, A \cap B^c, A \cup B^c, A^c \cap B, A^c \cup B\}$.

3.3 Borel Sets

(i) The **Borel sets** on $\mathbb{R}$ are sets obtained from countable unions, intersections, and complements of open sets.

(ii) The **Borel σ-algebra** $\mathcal{B}(\mathbb{R})$ is the smallest σ-algebra on $\mathbb{R}$ containing all open sets (equivalently, all closed sets).

The Borel σ-algebra can equivalently be generated by:

- All open sets, or all closed sets.
- All open intervals (a, b), or all closed intervals $[a, b]$.
- Sets of the form $(-\infty, a]$ for $a \in \mathbb{Q}$.
- Half-open intervals $(a, b]$, since $(a, b) = \bigcup_{n=1}^{\infty}(a, b - \frac{1}{n}]$.

3.4 Probability Measure

Let Ω be a non-empty set and $\mathcal{F}$ a σ-algebra on Ω. A **probability measure** is a function $\mathbb{P} : \mathcal{F} \to [0, 1]$ satisfying:

(i) $\mathbb{P}(\Omega) = 1$.

(ii) **Countable additivity:** For any sequence of pairwise disjoint events $A_1, A_2, \ldots \in \mathcal{F}$,

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n).$$

The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is called a **probability space**.

Remark 3.6. Countable additivity implies **finite additivity**: for pairwise disjoint $A_1, \ldots, A_n \in \mathcal{F}$,

$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} \mathbb{P}(A_i).$$

3.5 Properties of Probability Measures

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $A, B \in \mathcal{F}$. Then:

(i) $\mathbb{P}(\emptyset) = 0$.

(ii) $\mathbb{P}(A) \leq 1$.

(iii) If $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$ (monotonicity).

(iv) $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.

(v) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$ (inclusion–exclusion).

(vi) $\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B)$.

(vii) $\mathbb{P}(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} \mathbb{P}(A_n)$ (sub-additivity / union bound).

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $A_n \in \mathcal{F}$. If $A_n \uparrow A$ (i.e. $A_n \subseteq A_{n+1}$ and $\bigcup_{n=1}^{\infty} A_n = A$), then

$$\mathbb{P}(A_n) \uparrow \mathbb{P}(A), \quad \text{i.e.,} \quad \lim_{n\to\infty} \mathbb{P}(A_n) = \mathbb{P}(A).$$

Proof. Define $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$ for $n \geq 2$. The sets $\{B_n\}$ are pairwise disjoint and $\bigcup_{n=1}^{\infty} B_n = A$. By countable additivity,

$$\mathbb{P}(A) = \sum_{j=1}^{\infty} \mathbb{P}(B_j) = \lim_{n\to\infty} \sum_{j=1}^{n} \mathbb{P}(B_j) = \lim_{n\to\infty} \mathbb{P}(A_n). \qquad \square$$

3.6 Probability on Finite and Countable Spaces

For a finite or countable sample space Ω, we take $\mathcal{F} = 2^{\Omega}$. The probability measure is fully characterised by its values on singletons: setting $p_\omega = \mathbb{P}(\{\omega\})$ for each $\omega \in \Omega$, the requirements are

$$p_\omega \geq 0 \quad \forall\, \omega, \qquad \sum_{\omega\in\Omega} p_\omega = 1.$$

For any event $A \subseteq \Omega$, $\mathbb{P}(A) = \sum_{\omega\in A} p_\omega$.

4 Random Variables

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A **random variable** is a function $X : \Omega \to \mathbb{R}$ such that for every Borel set $B \in \mathcal{B}(\mathbb{R})$,

$$\{\omega \in \Omega : X(\omega) \in B\} \equiv X^{-1}(B) \in \mathcal{F}.$$

This condition is called **$\mathcal{F}$-measurability**.

Example 4.1 (Coin Toss)**.** Let $\Omega = \{H, T\}$ and $\mathcal{F} = \{\emptyset, \Omega, \{H\}, \{T\}\}$. Define $X(H) = 1$ and $X(T) = 0$. For any Borel set B:

$$X^{-1}(B) = \begin{cases} \{H\} & \text{if } 1 \in B,\ 0 \notin B, \\ \{T\} & \text{if } 0 \in B,\ 1 \notin B, \\ \emptyset & \text{if } 0, 1 \notin B, \\ \Omega & \text{if } 0, 1 \in B. \end{cases}$$

Each case belongs to $\mathcal{F}$, so X is a random variable.

Example 4.2 (Non-measurable function)**.** Let $\Omega = \{1, 2, 3, 4\}$ and $\mathcal{F} = \{\emptyset, \Omega, \{1\}, \{2, 3, 4\}\}$. The function $X(\omega) = 1 + \omega$ is **not** a random variable with respect to $\mathcal{F}$, since

$$X^{-1}(\{5\}) = \{\omega : 1 + \omega = 5\} = \{4\} \notin \mathcal{F}.$$

4.1 σ-Algebra Generated by a Random Variable

Let X be a random variable on $(\Omega, \mathcal{F}, \mathbb{P})$. The **$\sigma$-algebra generated by** X is

$$\sigma(X) := \{X^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\} = \{\{\omega : X(\omega) \in B\} : B \in \mathcal{B}(\mathbb{R})\}.$$

It is the smallest σ-algebra that makes X measurable.

Example 4.3. For $X = \mathbf{1}_{[0,\frac{1}{2}]}$ over $\Omega = [0, 1]$:

$$\sigma(X) = \left\{\emptyset,\ [0, 1],\ [0, \tfrac{1}{2}],\ (\tfrac{1}{2}, 1]\right\}.$$

Let $X_i : \Omega \to \mathbb{R}$, $i \in I$, be a collection of random variables. The **σ-algebra generated by the family** is

$$\sigma(X_i,\ i \in I) = \sigma\left(\bigcup_{i \in I} \sigma(X_i)\right).$$

Example 4.4. Let X_1, X_2 record the number of heads in two independent coin tosses, $\Omega = \{HH, HT, TH, TT\}$. Then

$$\begin{aligned}\sigma(X_1) &= \big\{\emptyset, \Omega, \{TT, TH\}, \{HH, HT\}\big\},\\ \sigma(X_2) &= \big\{\emptyset, \Omega, \{HT, TT\}, \{HH, TH\}\big\},\\ \sigma(X_1, X_2) &= \sigma\big(\{TT, TH\}, \{HH, HT\}, \{HT, TT\}, \{HH, TH\}\big).\end{aligned}$$

5 Expectation and Conditional Expectation

5.1 The Lebesgue Integral and Expectation

For a random variable X on $(\Omega, \mathcal{F}, \mathbb{P})$, the **expectation** (or **expected value**) is defined via the Lebesgue integral:

$$\mathbb{E}[X] = \int_\Omega X \, d\mathbb{P}.$$

The Lebesgue integral partitions the *range* of f (horizontal slices), while the Riemann integral partitions the *domain* (vertical slices). The Lebesgue approach is more general and handles discontinuous functions far better, making it essential for rigorous probability theory.

Let X, Y be integrable random variables and h a convex function. Then:

(i) **Linearity:** $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$.

(ii) **Positivity:** If $X \geq 0$ a.s., then $\mathbb{E}[X] \geq 0$.

(iii) **Jensen's Inequality:** $h(\mathbb{E}[X]) \leq \mathbb{E}[h(X)]$.

(iv) $\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A)$ for any event $A \in \mathcal{F}$.

Example 5.1 (Pascal's Wager — First Use of Expectation in Decision-Making)**.** Pascal argued that a rational person should believe in God. If we model the gain X of a believer as

$$X = \begin{cases} +\infty & \text{with probability } \frac{1}{2} \quad \text{(Heaven)},\\ -M & \text{with probability } \frac{1}{2} \quad \text{(finite loss)},\end{cases}$$

then the expected gain is

$$\mathbb{E}[X] = \infty \cdot \tfrac{1}{2} - M \cdot \tfrac{1}{2} = +\infty.$$

Therefore, by expected value reasoning, belief in God is always the rational choice.

5.2 Conditional Expectation

For a random variable X and an event B with $\mathbb{P}(B) \neq 0$, the **conditional expectation of X given B** is:

$$\mathbb{E}[X \mid B] = \frac{\mathbb{E}[\mathbf{1}_B X]}{\mathbb{P}(B)}.$$

For a random variable X and a discrete random variable Y taking values $y_1, y_2, \ldots$, the **conditional expectation** $\mathbb{E}[X \mid Y]$ is the random variable defined by:

$$\mathbb{E}[X \mid Y](\omega) = \mathbb{E}[X \mid \{Y = y_i\}] \quad \text{for every } \omega \in \{Y = y_i\}.$$

For a sub-σ-algebra $\mathcal{G} \subseteq \mathcal{F}$, the conditional expectation $\mathbb{E}[X \mid \mathcal{G}]$ is the (a.s. unique) $\mathcal{G}$-measurable random variable satisfying:

$$\int_G \mathbb{E}[X \mid \mathcal{G}]\, d\mathbb{P} = \int_G X\, d\mathbb{P} \quad \text{for all } G \in \mathcal{G}.$$

Intuitively, $\mathbb{E}[X \mid \mathcal{G}]$ is the "best prediction" of X given the information in $\mathcal{G}$.

6 Stochastic Processes

A **stochastic process** is a collection of random variables

$$\{X(t),\, t \in T\} = \{X(t, \omega),\, t \in T,\, \omega \in \Omega\}$$

defined on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$, where $T \subseteq [0, \infty)$ is the index set (time).

- **Fixed $t = t_0$:** $X(t_0, \cdot) : \Omega \to \mathbb{R}$ is an ordinary random variable.
- **Fixed $\omega = \omega_0$:** $X(\cdot, \omega_0) : T \to \mathbb{R}$ is a deterministic function of time, called a **sample path**, **trajectory**, or **realisation**.

Example 6.1. Prominent stochastic processes in finance include: Brownian motion, Poisson process, fractional Brownian motion, Gaussian processes, Markov processes, and Lévy processes.

7 Brownian Motion

7.1 Definition and Basic Properties

A stochastic process $\{B(t),\, t \geq 0\}$ is a **standard Brownian motion** if:

(i) $B(0) = 0$ almost surely.

(ii) B has **independent increments**: for $0 \leq t_0 < t_1 < \cdots < t_n$, the increments $B(t_1) - B(t_0),\, B(t_2) - B(t_1),\, \ldots,\, B(t_n) - B(t_{n-1})$ are independent.

(iii) B has **stationary increments**: $B(t) - B(s) \sim \mathcal{N}(0, t-s)$ for all $0 \leq s < t$.

(iv) Sample paths $t \mapsto B(t, \omega)$ are continuous (a.s.).

(i) $B(t) \sim \mathcal{N}(0, t)$, so $\mathbb{E}[B(t)] = 0$ and $\mathrm{Var}(B(t)) = t$.

(ii) **Covariance structure:** For $0 \leq s \leq t$,

$$\mathrm{Cov}(B(s), B(t)) = \min(s, t) = s.$$

(iii) Brownian motion is a **Martingale** (proved in Section 9).

(iv) The increments $B(t_1), B(t_2), \ldots, B(t_m)$ are **jointly normally distributed** with covariance matrix

$$C = \begin{pmatrix} t_1 & t_1 & \cdots & t_1 \\ t_1 & t_2 & \cdots & t_2 \\ \vdots & \vdots & \ddots & \vdots \\ t_1 & t_2 & \cdots & t_m \end{pmatrix}, \quad \text{i.e.,} \quad C_{ij} = \min(t_i, t_j).$$

7.2 Historical Context

- **Robert Brown (1827):** Observed erratic motion of pollen particles in water.
- **Albert Einstein (1905):** Provided the first mathematical explanation, connecting random molecular collisions to the diffusion equation.
- **Norbert Wiener (1926):** Constructed a rigorous mathematical model, now called the Wiener process.

7.3 Brownian Motion with Drift

Brownian motion with drift is defined by

$$B^\mu(t) = \mu t + \sigma B(t), \quad \mu \in \mathbb{R},\ \sigma > 0,$$

where $B(t)$ is a standard Brownian motion.

(i) $B^\mu(t) \sim \mathcal{N}(\mu t, \sigma^2 t)$, so $\mathbb{E}[B^\mu(t)] = \mu t$ and $\mathrm{Var}(B^\mu(t)) = \sigma^2 t$.

(ii) $\mathrm{Cov}(B^\mu(s), B^\mu(t)) = \sigma^2 \min(s,t)$.

(iii) $B^\mu(t)$ is **not a martingale** (unless $\mu = 0$).

(iv) $B^\mu(t)$ is a **submartingale** if $\mu > 0$; a **supermartingale** if $\mu < 0$.

The Brownian motion with drift satisfies the stochastic differential equation (SDE):

$$dB^\mu(t) = \mu\, dt + \sigma\, dB(t).$$

7.3.1 Simulating Brownian Motion

1. Discretise $[0,T]$ into $N = T/\Delta t$ uniform steps of length Δt.
2. Generate N independent normal random variables $\Delta B_k \sim \mathcal{N}(\mu\Delta t,\, \sigma^2\Delta t)$.
3. Compute $B^\mu(T) = \sum_{k=0}^{N-1} \Delta B_k$ (cumulative sum).

8 Filtered Probability Spaces

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A **filtration** is an increasing family $(\mathcal{F}_t)_{t\geq 0}$ of sub-σ-algebras of $\mathcal{F}$:

$$s \leq t \implies \mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}.$$

The quadruple $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ is called a **filtered probability space**.

The σ-algebra $\mathcal{F}_t$ represents the *information available up to time t*. As t increases, information accumulates and $\mathcal{F}_t$ grows. Formally, $\mathcal{F}_t$ contains any question that could possibly be asked and answered based on observations up to time t.

For a stochastic process $\{X_t\}$, the **natural filtration** is

$$\mathcal{F}_t^X = \sigma(X_s : 0 \leq s \leq t), \quad t \geq 0.$$

It is the smallest filtration to which X is adapted.

A stochastic process $\{X_t\}$ is **adapted** to the filtration $(\mathcal{F}_t)$ if, for each $t \geq 0$, the random variable X_t is $\mathcal{F}_t$-measurable. Every process is adapted to its own natural filtration.

9 Martingales

Martingales are one of the central objects in probability theory and mathematical finance. They formalise the notion of a *fair game*.

9.1 Discrete-Time Martingales

A sequence of random variables $X_1, X_2, X_3, \ldots$ is a **(discrete-time) martingale** with respect to a filtration $(\mathcal{F}_n)$ if, for all $n \geq 1$:

(i) $\mathbb{E}[|X_n|] < \infty$ (integrability).

(ii) $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n$ a.s. (fair game condition).

Taking $\mathcal{F}_n = \sigma(X_1, \ldots, X_n)$, condition (ii) reads $\mathbb{E}[X_{n+1} \mid X_1, \ldots, X_n] = X_n$.

Example 9.1 (Fair Gambling Game)**.** A gambler plays a fair game: he wins or loses Rs 100 with equal probability $\frac{1}{2}$. Let S_n denote his wealth after n rounds, with $X_k \in \{+100, -100\}$ being the outcome of round k. Then $S_n = S_{n-1} + X_n$ and:

$$\mathbb{E}[S_n \mid X_1, \ldots, X_{n-1}] = S_{n-1} + \mathbb{E}[X_n] = S_{n-1} + \tfrac{1}{2}(100) + \tfrac{1}{2}(-100) = S_{n-1}.$$

Hence S_n is a martingale.

9.2 Continuous-Time Martingales

A stochastic process $\{X_t,\, t \geq 0\}$ is a **martingale** with respect to $(\mathcal{F}_t)$ if, for all $0 \leq s < t$:

(i) $\mathbb{E}[|X_t|] < \infty$.

(ii) $\mathbb{E}[X_t \mid \mathcal{F}_s] = X_s$ a.s.

A process $\{X_t\}$ is a:

- **Submartingale** if $\mathbb{E}[X_t \mid \mathcal{F}_s] \geq X_s$ for $s < t$ (expected value is non-decreasing).
- **Supermartingale** if $\mathbb{E}[X_t \mid \mathcal{F}_s] \leq X_s$ for $s < t$ (expected value is non-increasing).

For a martingale: $\mathbb{E}[X_t] = \mathbb{E}[X_0]$ for all $t \geq 0$. For a submartingale: $\mathbb{E}[X_t] \geq \mathbb{E}[X_0]$. For a supermartingale: $\mathbb{E}[X_t] \leq \mathbb{E}[X_0]$.

9.3 Important Martingale Examples

9.3.1 Brownian Motion is a Martingale

Proposition 9.2. *Standard Brownian motion $\{B(t)\}$ is a martingale with respect to its natural filtration $\mathcal{F}_t^B$.*

Proof. First, $\mathbb{E}[|B(t)|] < \infty$ since $B(t) \sim \mathcal{N}(0, t)$. For $s < t$:

$$\mathbb{E}[B(t) \mid \mathcal{F}_s^B] = \mathbb{E}[(B(t) - B(s)) + B(s) \mid \mathcal{F}_s^B] = \underbrace{\mathbb{E}[B(t) - B(s)]}_{= 0, \text{ by stationarity}} + B(s) = B(s). \quad \square$$

9.3.2 Square Martingale

Proposition 9.3. *The process $M(t) = B(t)^2 - t$ is a martingale with respect to $\mathcal{F}_t^B$.*

Proof. We compute $\mathbb{E}[B(t)^2 - t \mid \mathcal{F}_s^B]$:

$$\begin{aligned}
\mathbb{E}[B(t)^2 - t \mid \mathcal{F}_s^B] &= \mathbb{E}[(B(t) - B(s) + B(s))^2 - t \mid \mathcal{F}_s^B] \\
&= \mathbb{E}[(B(t) - B(s))^2 + 2B(s)(B(t) - B(s)) + B(s)^2 - t \mid \mathcal{F}_s^B] \\
&= (t - s) + 2B(s) \cdot 0 + B(s)^2 - t = B(s)^2 - s. \qquad \square
\end{aligned}$$

9.3.3 Exponential Martingale

Proposition 9.4. *The process $Z(t) = e^{\sigma B(t) - \frac{1}{2}\sigma^2 t}$ is a martingale with respect to $\mathcal{F}_t^B$.*

Proof. For $X \sim \mathcal{N}(\mu, \sigma^2)$, the moment generating function is $\mathbb{E}[e^{uX}] = e^{\mu u + \frac{1}{2}u^2\sigma^2}$. In particular, for $B(t) - B(s) \sim \mathcal{N}(0, t - s)$:

$$\mathbb{E}\left[e^{\sigma(B(t) - B(s))}\right] = e^{\frac{1}{2}\sigma^2(t-s)}.$$

Therefore,

$$\begin{aligned}
\mathbb{E}[Z(t) \mid \mathcal{F}_s^B] &= e^{-\frac{1}{2}\sigma^2 t}\, \mathbb{E}\left[e^{\sigma B(t)} \mid \mathcal{F}_s^B\right] \\
&= e^{-\frac{1}{2}\sigma^2 t}\, e^{\sigma B(s)}\, \mathbb{E}\left[e^{\sigma(B(t) - B(s))}\right] \\
&= e^{-\frac{1}{2}\sigma^2 t}\, e^{\sigma B(s)}\, e^{\frac{1}{2}\sigma^2(t-s)} = e^{\sigma B(s) - \frac{1}{2}\sigma^2 s} = Z(s). \qquad \square
\end{aligned}$$

The exponential martingale $Z(t) = e^{\sigma B(t) - \frac{1}{2}\sigma^2 t}$ arises directly in the Black–Scholes model. A geometric Brownian motion stock price $S(t) = S(0)e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma B(t)}$ is a submartingale under the physical measure $\mathbb{P}$, but becomes a martingale under the risk-neutral measure $\mathbb{Q}$ via a change of measure given by the exponential martingale.

9.3.4 Poisson Process

Proposition 9.5. *The standard Poisson process $N(t)$ with intensity $\lambda > 0$ is* not *a martingale. However, the* ***compensated Poisson process*** *$M(t) = N(t) - \lambda t$ is a martingale.*

Proof. Using independent increments of N:

$$\mathbb{E}[N(t) - N(s) \mid \mathcal{F}_s^N] = \mathbb{E}[N(t) - N(s)] = \lambda(t-s) \neq 0,$$

so $N(t)$ is a submartingale (not a martingale). For the compensated process:

$$\mathbb{E}[M(t) - M(s) \mid \mathcal{F}_s^N] = \mathbb{E}[N(t) - N(s) - \lambda(t-s)] = \lambda(t-s) - \lambda(t-s) = 0,$$

hence $M(t)$ is a martingale. □

10 Introduction to Stochastic Calculus

10.1 Motivation: Stock Prices

Empirical stock price data (e.g. Google, Facebook, Walmart, State Bank of India) exhibit several stylised facts:

- Continuous but non-differentiable paths.
- Variance (volatility) proportional to the time step.
- Fat-tailed returns over short periods.

These features motivate modelling stock prices as geometric Brownian motion, requiring a non-standard calculus — **Itô calculus**.

10.2 Geometric Brownian Motion

The standard model for a risky asset price $S(t)$ is:

$$dS(t) = \mu S(t)\,dt + \sigma S(t)\,dB(t),$$

which has the closed-form solution

$$S(t) = S(0)\exp\left(\left(\mu - \tfrac{1}{2}\sigma^2\right)t + \sigma B(t)\right).$$

10.3 Itô's Lemma

Itô's Lemma

Let $f(t,x)$ be a twice continuously differentiable function and $\{X_t\}$ satisfy

$$dX_t = \mu_t\,dt + \sigma_t\,dB_t.$$

Then:

$$df(t, X_t) = \frac{\partial f}{\partial t}\, dt + \frac{\partial f}{\partial x}\, dX_t + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(dX_t)^2,$$

where $(dB_t)^2 = dt$, $(dt)^2 = 0$, $dt \cdot dB_t = 0$ (Itô's multiplication table). Explicitly:

$$df(t, X_t) = \left(\frac{\partial f}{\partial t} + \mu_t \frac{\partial f}{\partial x} + \frac{1}{2}\sigma_t^2 \frac{\partial^2 f}{\partial x^2}\right) dt + \sigma_t \frac{\partial f}{\partial x}\, dB_t.$$

In classical calculus, $(dx)^2 \to 0$ is negligible. In Itô calculus, $(dB_t)^2 = dt \neq 0$ is of first order, giving rise to the extra term $\frac{1}{2}\sigma^2 f_{xx}$. This extra term is crucial in the derivation of the Black–Scholes PDE.

10.4 Pioneers in Mathematical Finance

- **Louis Bachelier (1900):** First to model stock prices as Brownian motion in his PhD thesis "Théorie de la Spéculation".
- **Kiyosi Itô (1944):** Developed the stochastic integral and Itô's lemma.
- **Black, Scholes & Merton (1973):** Derived the Black–Scholes option pricing formula, earning Scholes and Merton the 1997 Nobel Prize.
- **Harry Markowitz (1952, Nobel 1990):** Founded modern portfolio theory using expected return and variance (mean–variance optimisation).

11 Summary of Key Definitions

Concept	**Definition / Characterisation**
Sample space Ω	Set of all possible outcomes
σ-algebra $\mathcal{F}$	Collection of events closed under complement and countable union
Probability measure $\mathbb{P}$	$\mathbb{P} : \mathcal{F} \to [0, 1]$ with $\mathbb{P}(\Omega) = 1$ and countable additivity
Random variable X	$\mathcal{F}$-measurable function $X : \Omega \to \mathbb{R}$
Expectation $\mathbb{E}[X]$	Lebesgue integral $\int_\Omega X\, d\mathbb{P}$
Filtration $(\mathcal{F}_t)$	Increasing family of sub-σ-algebras (information flow)
Brownian motion $B(t)$	Continuous, $\mathcal{N}(0, t)$-distributed, independent stationary increments
Martingale	$\mathbb{E}[X_t \mid \mathcal{F}_s] = X_s$ for $s < t$ (fair game)

12 References

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